

BENLYSTA

Belimumab

QUALITATIVE AND QUANTITATIVE COMPOSITION

120 mg vial

Each vial contains 120 mg belimumab (80 mg/mL after reconstitution).

Belimumab is a human IgG1 λ monoclonal antibody specific for soluble human B Lymphocyte Stimulator protein (BLyS). Belimumab is produced by recombinant DNA technology in a mammalian cell expression system.

White to off-white powder

Powder for concentrate for solution for infusion.

CLINICAL PARTICULARS

Indications

BENLYSTA is indicated for:

- reducing disease activity in adult patients with active autoantibody positive systemic lupus erythematosus (SLE) who are receiving standard therapy.
- treatment of active lupus nephritis in adult patients in combination with background immunosuppressive therapies (*see Dosage and Administration*).

BENLYSTA has not been studied and is therefore not recommended in patients with severe active central nervous system lupus.

Dosage and Administration

BENLYSTA is administered intravenously by infusion, and must be reconstituted and diluted prior to administration (*see Use and Handling*).

BENLYSTA treatment should be initiated and supervised by a qualified physician experienced in the diagnosis and treatment of SLE.

BENLYSTA should be administered by a healthcare professional prepared to treat hypersensitivity reactions including anaphylaxis.

BENLYSTA should be infused over a 1-hour period.

BENLYSTA must not be administered as an intravenous push or bolus.

The infusion rate may be slowed or interrupted if the patient develops an infusion reaction. The infusion must be discontinued immediately if the patient experiences a potentially life-threatening adverse reaction (*see Contraindications, Warnings and Precautions*).

Patients should be monitored during and for an appropriate period of time after administration of *BENLYSTA* (see *Warnings and Precautions, Adverse Reactions*).

Premedication for intravenous infusion

Premedication with an oral antihistamine, with or without an antipyretic, may be administered before the infusion of *BENLYSTA* (see *Warnings and Precautions, Clinical Studies*).

Adults

SLE

The recommended dosage regimen is 10 mg/kg on Days 0, 14 and 28, and at 4-week intervals thereafter. Discontinuation of treatment with *BENLYSTA* should be considered if there is no improvement in disease control after 6 months of treatment.

Lupus nephritis

The recommended dosage regimen is 10 mg/kg on Days 0, 14 and 28, and at 4-week intervals thereafter.

In patients with active lupus nephritis, *BENLYSTA* should be used in combination with corticosteroids and mycophenolate or cyclophosphamide for induction, or mycophenolate or azathioprine for maintenance.

Children

SLE

BENLYSTA has not been studied in patients less than 18 years of age. There are no data on the safety and efficacy of *BENLYSTA* in this age group.

Lupus nephritis

The safety and efficacy of *BENLYSTA* in children and adolescents aged below 18 years have not been established, therefore, *BENLYSTA* is not recommended for use in children and adolescents with active lupus nephritis.

Elderly

The efficacy and safety of *BENLYSTA* in the elderly has not been established. Data on patients >65 years are limited to < 1.6% of the studied population. Therefore, the use of *BENLYSTA* in elderly patients is not recommended unless the benefits are expected to outweigh the risks. In case administration of *BENLYSTA* to elderly patient is deemed necessary, dose adjustment is not required (see *Pharmacokinetics - Special Patient Groups*).

Renal impairment

No formal studies of *BENLYSTA* have been performed in patients with renal impairment.

BENLYSTA has been studied in a limited number of SLE patients with renal impairment. Dosage adjustment is not required in patients with renal impairment (see *Pharmacokinetics - Special Patient Groups*).

Hepatic impairment

No formal studies of *BENLYSTA* have been performed in patients with hepatic impairment. However, patients with hepatic impairment are unlikely to require dose modification (see *Pharmacokinetics - Special Patient Groups*).

Contraindications

BENLYSTA is contraindicated in patients who have demonstrated anaphylaxis to *BENLYSTA*.

Warnings and Precautions

Traceability

In order to improve traceability of biological medicinal products, the tradename and the batch number of the administered product should be clearly recorded.

BENLYSTA has not been studied in the following adult and paediatric patient groups, and is not recommended in:

- severe active central nervous system lupus (see section 5.1)
- HIV
- a history of, or current, hepatitis B or C
- hypogammaglobulinaemia (IgG < 400 mg/dL) or IgA deficiency (IgA < 10 mg/dL)
- a history of major organ transplant or hematopoietic stem cell /marrow transplant or renal transplant.

Concomitant use with B cell targeted therapy

Available data do not support the co-administration of rituximab with *BENLYSTA* in patients with SLE (see *Pharmacodynamic*). Caution should be exercised if *BENLYSTA* is co-administered with other B cell targeted therapy.

Infusion reactions and hypersensitivity

Administration of *BENLYSTA* may result in hypersensitivity reactions and infusion reactions which can be severe, and fatal. In the event of a severe reaction, *BENLYSTA* administration must be interrupted and appropriate medical therapy administered (see *Dosage and Administration*). The risk of hypersensitivity reactions is greatest with the first two infusions; however the risk should be considered for every infusion administered. Patients with a history of multiple drug allergies or significant hypersensitivity may be at increased risk.

Premedication including an antihistamine, with or without an antipyretic, may be administered before the infusion of *BENLYSTA*. There is insufficient knowledge to determine whether premedication could diminish the frequency or severity of infusion reactions.

In clinical studies, serious infusion and hypersensitivity reactions affected approximately 0.9 % of adult patients, and included anaphylactic reaction, bradycardia, hypotension, angioedema, and dyspnoea. Infusion reactions occurred more frequently during the first two infusions and tended to decrease with subsequent infusions (see *Adverse Reactions*). Patients have been reported to develop symptoms of acute hypersensitivity several hours after the infusion has been administered. Recurrence of clinically significant reactions after initial appropriate

treatment of symptoms has also been observed (*see Dosage and Administration and Adverse Reactions*). Therefore, BENLYSTA should be administered in an environment where resources for managing such reactions are immediately available. Patients should remain under clinical supervision for a prolonged period of time (for several hours), following at least the first 2 infusions, taking into account the possibility of a late onset reaction. Patients should be advised that hypersensitivity reactions are possible, on the day of, or several days after infusion, and be informed of potential signs and symptoms and the possibility of recurrence. Patients should be instructed to seek immediate medical attention if they experience any of these symptoms. The package leaflet should be provided to the patient each time BENLYSTA is administered (*see section Dosage and Administration*).

Delayed-type, non-acute hypersensitivity reactions have also been observed and included symptoms such as rash, nausea, fatigue, myalgia, headache, and facial oedema.

Infections

The mechanism of action of belimumab could increase the risk for the development of infections in adults and children with lupus, including opportunistic infections, and younger children may be at increased risk. In controlled clinical studies, the incidence of serious infections was similar across the BENLYSTA and placebo groups; however, fatal infections (e.g. pneumonia and sepsis) occurred more frequently in patients receiving BENLYSTA compared with placebo (*See Adverse Reactions*). Pneumococcal vaccination should be considered before initiating BENLYSTA treatment. BENLYSTA should not be initiated in patients with active serious infections (including serious chronic infections). Physicians should exercise caution and carefully assess if the benefits are expected to outweigh the risks when considering the use of BENLYSTA in patients with a history of recurrent infection. Physicians should advise patients to contact their health care provider if they develop symptoms of an infection. Patients who develop an infection while undergoing treatment with BENLYSTA should be monitored closely and careful consideration given to interrupting immunosuppressant therapy including BENLYSTA until the infection is resolved. The risk of using BENLYSTA in patients with active or latent tuberculosis is unknown.

Depression and suicidality

In controlled clinical intravenous and subcutaneous studies, psychiatric disorders (depression, suicidal ideation and behaviour including suicides) have been reported more frequently in patients receiving BENLYSTA (*See Adverse Reactions*). Physicians should assess the risk of depression and suicide considering the patient's medical history and current psychiatric status before treatment with BENLYSTA and continue to monitor patients during treatment. Physicians should advise patients (and caregivers where appropriate) to contact their health care provider about new or worsening psychiatric symptoms. In patients who experience such symptoms, treatment discontinuation should be considered.

Progressive multifocal leukoencephalopathy

Progressive multifocal leukoencephalopathy (PML) has been reported with BENLYSTA treatment for SLE. Physicians should be particularly alert to symptoms suggestive of PML that patients may not notice (e.g., cognitive, neurological or psychiatric symptoms or signs). Patients should be monitored for any of these new or worsening symptoms or signs, and if such symptoms/signs occur, referral to a neurologist and appropriate diagnostic measures for PML should be considered as clinical indicated. If PML is suspected, immunosuppressant therapy, including BENLYSTA, must be suspended until PML has been excluded. If PML is confirmed, immunosuppressant therapy, including BENLYSTA, must be discontinued.

Immunisation

Live vaccines should not be given for 30 days before, or concurrently with BENLYSTA, as clinical safety has not been established. No data are available on the secondary transmission of infection from persons receiving live vaccines to patients receiving BENLYSTA.

Because of its mechanism of action, belimumab may interfere with the response to immunisations. However, in a small study evaluating the response to a 23-valent pneumococcal vaccine, overall immune responses to the different serotypes were similar in SLE patients receiving BENLYSTA compared with those receiving standard immunosuppressive treatment at the time of vaccination. There are insufficient data to draw conclusions regarding response to other vaccines.

Limited data suggest that BENLYSTA does not significantly affect the ability to maintain a protective immune response to immunisations received prior to administration of BENLYSTA. In a substudy, a small group of patients who had previously received either tetanus, pneumococcal or influenza vaccinations were found to maintain protective titres after treatment with BENLYSTA.

Malignancies and lymphoproliferative disorders

Immunomodulatory medicinal products, including BENLYSTA, may increase the risk of malignancy. Caution should be exercised when considering BENLYSTA therapy for patients with a history of malignancy or when considering continuing treatment in patients who develop malignancy. Patients with malignant neoplasm within the last 5 years have not been studied, with the exception of those with basal or squamous cell cancers of the skin, or cancer of the uterine cervix, that has been fully excised or adequately treated.

Sodium content

This medicinal product contains less than 1 mmol sodium (23 mg) per dose, i.e. essentially 'sodium-free'.

Interactions

No drug interaction studies have been conducted with *BENLYSTA*.

In clinical trials, concomitant administration of mycophenolate mofetil, cyclophosphamide, azathioprine, hydroxychloroquine, methotrexate, non-steroidal anti-inflammatory medications, aspirin, and HMG CoA reductase inhibitors had no significant effect on belimumab exposures (see *Pharmacokinetics*).

Pregnancy and Lactation

Fertility

There are no data on the effects of *BENLYSTA* on human fertility. Effects on male and female fertility have not been evaluated in animal studies (see *Non-Clinical Information*).

Pregnancy

There are limited data on the use of *BENLYSTA* in pregnant women. Post-marketing data from a prospective pregnancy registry have collected pregnancy information in women exposed to belimumab. Due to the small sample size achieved, no definitive conclusions from this registry can be made regarding a potential risk of birth defects following exposure to belimumab. Immunoglobulin G (IgG) antibodies, including belimumab, can cross the placenta. *BENLYSTA*

should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

If prevention of pregnancy is warranted, women of childbearing potential should use adequate contraception while using *BENLYSTA* and for at least four months after the last *BENLYSTA* treatment.

Animal studies did not indicate direct or indirect harmful effects with respect to maternal toxicity, pregnancy or embryofoetal development. Treatment-related findings were limited to reversible reductions in B cells in infant monkeys (*see Non-Clinical Information*). Monitor infants of treated mothers for B-cell reduction and depending upon the results, consider delaying infant vaccination with live viral vaccines. B-cell reduction in infants may also interfere with the response to immunisations (*see Warnings and Precautions*).

Lactation

The safety of *BENLYSTA* for use during lactation has not been established. There are no data regarding the excretion of belimumab in human milk, or systemic absorption of belimumab after ingestion. Although belimumab was excreted into the milk of cynomolgous monkeys, published literature suggests that human neonatal and infant consumption of breast milk does not result in clinically significant absorption of maternal IgG antibodies into circulation.

It is recommended that a decision should be made about *BENLYSTA* therapy in breast-feeding mothers, taking into account the importance of breast-feeding to the infant and the importance of the drug to the mother, and any potential adverse effects on the breastfed child from belimumab or from the underlying maternal condition.

Effects on Ability to Drive and Use Machines

There have been no studies to investigate the effect of *BENLYSTA* on driving performance or the ability to operate machinery. No detrimental effects on such activities are predicted from the pharmacology of belimumab.

The clinical status of the patient and the safety profile of *BENLYSTA* should be borne in mind when considering the patient's ability to perform tasks that require judgement, motor or cognitive skills.

Adverse Reactions

Adults

The safety of *BENLYSTA* in patients with SLE has been evaluated in three pre-registration placebo-controlled intravenous studies and one subsequent regional placebo-controlled intravenous study, one placebo-controlled subcutaneous study, and two post-marketing placebo-controlled intravenous studies; the safety in patients with active lupus nephritis has been evaluated in one placebo-controlled intravenous study.

The data described below reflect exposure in 674 patients with SLE from the three pre-registration clinical studies and 470 patients in the subsequent placebo-controlled study administered *BENLYSTA* intravenously (10 mg/kg over a 1-hour period on Days 0, 14, 28, and then every 28 days up to 52 weeks), and 556 patients with SLE administered *BENLYSTA* subcutaneously (200 mg once weekly up to 52 weeks). The safety data presented include data beyond Week 52 in some patients with SLE. The data reflect additional exposure in 224 patients with active lupus nephritis who received *BENLYSTA* intravenously (10 mg/kg for up to 104 weeks). Data from post-marketing reports are also included.

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The majority of patients were also receiving one or more of the following concomitant treatments for SLE: corticosteroids, immunomodulatory agents, anti-malarials, non-steroidal anti-inflammatory drugs. Adverse reactions are listed below by MedDRA body system organ class and by frequency. The frequency categories used are:

Very common $\geq 1/10$

Common $\geq 1/100$ and $< 1/10$

Uncommon $\geq 1/1,000$ and $< 1/100$

MedDRA SOC	Very common:	Common:	Uncommon:
<i>Infections and infestations</i>	Infections		
<i>Immune System Disorders</i>		Hypersensitivity reaction*	Anaphylactic reaction Angioedema
<i>Psychiatric disorders</i>		Depression	Suicidal ideation Suicidal behaviour
<i>Skin and Subcutaneous Tissue Disorders</i>		Rash Urticaria	
<i>General Disorders and Administration Site Conditions</i>		Pyrexia Infusion related systemic reactions*	

* 'Hypersensitivity reaction' covers a group of terms, including anaphylaxis, and can manifest as a range of symptoms including hypotension, angioedema, urticaria or other rash, pruritus, and dyspnoea. 'Infusion-related reaction' covers a group of terms and can manifest as a range of symptoms including bradycardia, myalgia, headache, rash, urticaria, pyrexia, hypotension, hypertension, dizziness, and arthralgia. Due to overlap in signs and symptoms, it is not possible to distinguish between hypersensitivity reactions and infusion reactions in all cases.

Hypersensitivity reactions: In the three intravenous SLE pre-registration studies, clinically significant hypersensitivity reactions associated with *BENLYSTA* and requiring permanent treatment discontinuation were reported in 0.4% of patients. These reactions were generally observed on the day of the infusion, and patients with a history of multiple drug allergies or significant hypersensitivity reactions may be at increased risk. Delay in the onset of acute hypersensitivity reactions for several hours after the infusion, and recurrence of clinically significant reactions after initial resolution of symptoms following appropriate treatment, have been observed. Delayed-type, non-acute hypersensitivity reactions have also been observed and included symptoms such as rash, nausea, fatigue, myalgia, headache, and facial oedema.

Infections: In the three intravenous SLE pre-registration clinical studies, the overall incidence of infections was 70% in the group receiving belimumab and 67% in the group receiving placebo. Infections occurring in at least 3% of patients receiving *BENLYSTA* and at least 1% more frequently than patients receiving placebo were nasopharyngitis, bronchitis, pharyngitis,

cystitis, and gastroenteritis viral. Serious infections occurred in 5% of patients receiving either belimumab or placebo; serious opportunistic infections accounted for <1% and 0% of these, respectively. Some infections were severe or fatal.

In a randomised (1:1), double-blind, placebo-controlled, 52-week, post-marketing SLE safety study (BEL115467) which assessed mortality and specific adverse events in adults, serious infections occurred in 3.7% of patients receiving *BENLYSTA* 10 mg/kg intravenously and in 4.1% of patients receiving placebo. Fatal infections occurred in 0.45% (9/2002) of patients receiving *BENLYSTA* and in 0.15% (3/2001) of patients receiving placebo, while the incidence of all-cause mortality was 0.50% (10/2002) in patients receiving belimumab and 0.40% (8/2001) in patients receiving placebo.

In the lupus nephritis study, patients were receiving a background of standard therapy (see Clinical Studies) and the overall incidence of infections was 82% in patients receiving *BENLYSTA* compared with 76% in patients receiving placebo. Serious infections occurred in 13.8% of patients receiving *BENLYSTA* and in 17.0% of patients receiving placebo. Fatal infections occurred in 0.9% (2/224) of patients receiving *BENLYSTA* and in 0.9% (2/224) of patients receiving placebo.

Psychiatric disorders: In the three pre-registration SLE intravenous clinical studies, serious psychiatric events were reported in 1.2% (8/674) of patients receiving *BENLYSTA* 10 mg/kg and 0.4% (3/675) of patients receiving placebo. Serious depression was reported in 0.6% (4/674) of patients receiving *BENLYSTA* 10 mg/kg and 0.3% (2/675) of patients receiving placebo. One suicide was reported in a patient receiving *BENLYSTA* 10 mg/kg (and one was reported in a patient receiving *BENLYSTA* 1 mg/kg); there were no reports in patients receiving placebo.

In a large post-marketing SLE study, serious psychiatric events were reported in 1.0% (20/2002) of patients receiving *BENLYSTA* and 0.3% (6/2001) of patients receiving placebo. Serious depression was reported in 0.3% (7/2002) of patients receiving *BENLYSTA* and <0.1% (1/2001) receiving placebo. The overall incidence of serious suicidal ideation or behaviour or self-injury without suicidal intent was 0.7% (15/2002) in the *BENLYSTA* group and 0.2% (5/2001) in the placebo group. On the Columbia-Suicide Severity Rating Scale (C-SSRS), 2.4% (48/1974) of patients receiving *BENLYSTA* reported suicidal ideation or behaviour compared with 2.0% (39/1988) of patients receiving placebo. No suicide was reported in either group.

The SLE intravenous studies above did not exclude patients with a history of psychiatric disorders.

In the subcutaneous SLE clinical study, which excluded patients with a history of psychiatric disorders, serious psychiatric events were reported in 0.2% (1/556) of patients receiving *BENLYSTA* and in no patients receiving placebo. There were no serious depression-related events or suicides reported in either group. On the C-SSRS, 1.3% (7/554) of patients receiving *BENLYSTA* reported suicidal ideation or behaviour and 0.7% (2/277) of patients receiving placebo.

Overdose

There is limited experience with overdosage of *BENLYSTA*. Adverse reactions reported in association with cases of overdose have been consistent with those expected for belimumab.

Two doses up to 20 mg/kg administered 21 days apart by intravenous infusion have been given to humans with no increase in incidence or severity of adverse reactions compared with doses

of 1, 4, or 10 mg/kg.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamics

ATC Code

Pharmacotherapeutic group: Immunosuppressants, monoclonal antibodies

ATC code: L04AG04

Mechanism of Action

B-Lymphocyte Stimulator (BLyS, also referred to as BAFF and TNFSF13), a member of the tumour necrosis factor (TNF) ligand family, inhibits B cell apoptosis and stimulates the differentiation of B cells into immunoglobulin-producing plasma cells. BLyS is overexpressed in patients with SLE leading to elevated plasma BLyS levels. There is a strong association between SLE disease activity (as assessed by the Safety of Estrogen in Lupus Erythematosus National Assessment-Systemic Lupus Erythematosus Disease Activity Index [SELENA-SLEDAI]) and plasma BLyS levels.

Belimumab is a human IgG1 λ monoclonal antibody that specifically binds to soluble human BLyS and inhibits its biological activity. Belimumab does not bind B cells directly, but by binding and neutralizing BLyS, belimumab inhibits the survival of B cells, including autoreactive B cells, and reduces the differentiation of B cells into immunoglobulin producing plasma cells.

Pharmacodynamic Effect

In adult patients with SLE, reductions in elevated levels of serum IgG and in anti-dsDNA antibodies were observed as early as Week 8 and continued to Week 52. In patients with hypergammaglobulinaemia at baseline, normalisation of IgG levels was observed by week 52 in 49% and 20% of patients receiving belimumab and placebo, respectively. In patients with anti-dsDNA antibodies at baseline, reductions in patients receiving belimumab were evident as early as Week 8, and by Week 52, 16% of patients treated with belimumab had converted to anti-dsDNA negative compared with 7% of the patients receiving placebo.

In patients with SLE with low complement levels at baseline, belimumab treatment resulted in increases in complement C3 and C4 which were seen as early as Week 4 and continued over time. By Week 52, levels of C3 and C4 had normalised in 38% and 44% of patients receiving belimumab compared with 17% and 19% of patients receiving placebo.

The target of belimumab, BLyS, is a critical cytokine for B cell survival, differentiation, and proliferation. Belimumab significantly reduced circulating B cells, naïve, activated, plasma, and the SLE B cell subset at Week 52. Reductions in naïve, plasma and short-lived plasma cells as well as the SLE B cell subset were observed as early as Week 8. Memory cells increased initially and slowly declined toward baseline levels by Week 52.

In a long-term uncontrolled extension SLE study, B cells (including naïve, activated, plasma cells and the SLE B cell subset) and IgG levels were followed for more than 7 years with ongoing treatment. A substantial and sustained decrease in various B cell subsets was observed leading to median reductions of 87% in naïve B cells, 67% in memory B cells, 99% in activated B cells, and 92% in plasma cells after more than 7 years of treatment. After about 7 years, a 28% median reduction in IgG levels was observed with 1.6% of subjects

experiencing a decrease in IgG levels to below 400 mg/dL. Over the course of the study, the reported incidence of AEs generally remained stable or declined.

In patients with active lupus nephritis, following treatment with belimumab or placebo, there was an increase in serum IgG levels which was associated with decreased proteinuria. Relative to placebo, smaller increases in serum IgG levels were observed in the belimumab group as expected with the known mechanism of belimumab. At Week 104, the median percent increase from baseline in IgG was 17% for belimumab and 37% for placebo. Reductions in autoantibodies, increases in complement, and reductions in circulating total B cells and B-cell subsets observed were consistent with the SLE studies.

Immunogenicity

In the two Phase III SLE studies with *BENLYSTA* administered intravenously in adult patients, 4 out of 563 (0.7%) patients in the 10 mg/kg group and 27 out of 559 (4.8%) patients in the 1 mg/kg group developed persistent anti-belimumab antibodies. The reported frequency for the 10 mg/kg group may underestimate the actual frequency due to lower assay sensitivity in the presence of high drug concentrations. Neutralising antibodies were detected in three patients receiving belimumab 1 mg/kg intravenously.

In a Phase III/IV study with *BENLYSTA* 10 mg/kg administered intravenously to black patients with SLE, two of the 321 (0.6%) patients developed anti-belimumab antibodies.

In a Phase III study with *BENLYSTA* 10 mg/kg administered intravenously to adult patients with active lupus nephritis, none of the 224 patients developed anti-belimumab antibodies.

The presence of anti-belimumab antibodies was relatively uncommon in patients treated with *BENLYSTA* and no definitive conclusions can be drawn regarding the effect of immunogenicity on belimumab pharmacokinetics due to low numbers of anti-belimumab antibody positive subjects.

Pharmacokinetics

SLE studies

The pharmacokinetic parameters below are based on population parameter estimates from 563 patients with SLE who received belimumab 10 mg/kg intravenously (Days 0, 14, 28, and then every 28 days up to 52 weeks) in the two Phase III studies in adults.

Absorption

Following intravenous administration, maximum serum concentrations ($C_{\max}$) of belimumab were generally observed at, or shortly after, the end of the infusion. The $C_{\max}$ was 313 µg/mL at steady-state based on simulating the concentration time profile using the typical parameter values of the population pharmacokinetic model.

Distribution

Belimumab was distributed to tissues with an overall volume of distribution of approximately 5 L.

Metabolism

Belimumab is a protein for which the expected metabolic pathway is degradation to small peptides and individual amino acids by widely distributed proteolytic enzymes. Classical

biotransformation studies have not been conducted.

Elimination

Serum belimumab concentrations declined in a bi-exponential manner, with a distribution half-life of 1.75 days and terminal half-life 19.4 days. The systemic clearance was 215 mL/day.

Lupus nephritis study

A population pharmacokinetic analysis was conducted in 224 adult patients with lupus nephritis who received *BENLYSTA* 10 mg/kg intravenously (Days 0, 14, 28, and then every 28 days up to 104 weeks). In patients with lupus nephritis, due to renal disease activity, belimumab clearance was initially higher than observed in SLE studies; however, after 24 weeks of treatment and throughout the remainder of the study, belimumab clearance and exposure were similar to that observed in adult patients with SLE who received *BENLYSTA* 10 mg/kg intravenously.

Transitioning from Intravenous to Subcutaneous Administration

SLE

Patients with SLE transitioning from 10 mg/kg intravenously every 4 weeks to 200 mg subcutaneously weekly using a 1 to 4 week switching interval had pre-dose belimumab serum concentrations at their first subcutaneous dose close to their eventual subcutaneous steady-state trough concentration (*see Dosage and Administration*). Based on simulations with population pharmacokinetic parameters, the steady-state average belimumab concentrations for 200 mg subcutaneous every week were similar to 10 mg/kg intravenous every 4 weeks.

Lupus nephritis

One to 2 weeks after completing the first 2 intravenous doses, patients with lupus nephritis transitioning from 10 mg/kg intravenously to 200 mg subcutaneously weekly, are predicted to have average belimumab serum concentrations similar to patients dosed with 10 mg/kg intravenously every 4 weeks based on population PK simulations (*see Dosage and Administration*).

Drug interactions

Concomitant use of mycophenolate mofetil, cyclophosphamide, azathioprine, methotrexate and hydroxychloroquine did not substantially influence belimumab pharmacokinetics based on the results of population pharmacokinetic analyses. Neither did a wide range of other co-medications (non-steroidal anti-inflammatory medications, aspirin, and HMG- CoA reductase inhibitors) significantly influence belimumab pharmacokinetics. Co-administration of steroids and ACE inhibitors resulted in a statistically significant increase of systemic clearance in the population pharmacokinetic analysis. However, these effects were not clinically meaningful as their magnitude was well within the range of normal variability of clearance.

Special Patient Groups

Elderly

Belimumab has been studied in a limited number of elderly patients. Age did not affect belimumab exposure in the population pharmacokinetic analyses. However, given the small number of subjects 65 years or older, an effect of age cannot be ruled out conclusively.

Children and adolescents

No pharmacokinetic data are available in paediatric patients.

Renal impairment

No formal studies were conducted to examine the effects of renal impairment on the pharmacokinetics of belimumab. During clinical development, belimumab was studied in a limited number of SLE patients with renal impairment (creatinine clearance less than 60 mL/min, including a small number with creatinine clearance less than 30 mL/min). Although proteinuria (greater than or equal to 2 g/day) increased belimumab clearance, and decreases in creatinine clearance decreased belimumab clearance, these effects were within the expected range of variability. Therefore, no dose adjustment is recommended for patients with renal impairment.

Hepatic impairment

No formal studies were conducted to examine the effects of hepatic impairment on the pharmacokinetics of belimumab. IgG1 molecules such as belimumab are catabolised by widely distributed proteolytic enzymes, which are not restricted to hepatic tissue; therefore, changes in hepatic function are unlikely to have any effect on the elimination of belimumab.

Other patient characteristics

There was no significant effect of gender, race or ethnicity on the pharmacokinetics of belimumab. The effects of body size on belimumab exposure are accounted for by weight normalised dosing.

Clinical Studies

Intravenous infusion in adult patients

SLE

The efficacy of *BENLYSTA* was evaluated in two randomised, double-blind, placebo-controlled Phase III studies in 1,684 patients with a clinical diagnosis of SLE according to the American College of Rheumatology (ACR) classification criteria. Eligible patients had active SLE disease, defined as a SELENA-SLEDAI score greater than or equal to 6 and positive anti-nuclear antibody (ANA or anti-dsDNA) test results (ANA titre greater than or equal to 1:80 and/or a positive anti-dsDNA [greater than or equal to 30 units/mL]) at screening. Patients were on a stable SLE treatment regimen (standard of care) consisting of any of the following (alone or in combination): corticosteroids, anti-malarials, NSAIDs or other immunosuppressives. Patients were excluded from the study if they had severe active central nervous system lupus or severe active lupus nephritis, had ever received treatment with any B cell targeted therapy, if they had received another biological investigational agent in the previous year, or if they had a positive response to testing for HIV antibody, hepatitis B surface antigen, or hepatitis C antibody. The two studies were similar in design except that Study 1 was a 76-week study and Study 2 was a 52-week study. Both studies had 52-week primary endpoints.

Study 1 (HGS1006-C1056) was conducted primarily in North America and Western Europe. The racial distribution was 70% white/Caucasian, 14% black/African American, 13% Alaska native or American Indian, and 3% Asian. Background medications included corticosteroids (76%), immunosuppressives (56%), and anti-malarials (63%).

Study 2 (HGS1006-C1057) was conducted in South America, Eastern Europe, Asia, and

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Australia. The racial distribution was 38% Asian, 26% white/Caucasian, 32% Alaska native or American Indian, and 4% black/African American. Background medications included corticosteroids (96%), immunosuppressives (42%), and anti-malarials (67%).

Patient median age across both studies was 37 years (range: 18 to 73 years), and the majority (94%) were female. At screening, patients were stratified by disease severity based on their SELENA-SLEDAI score (less than or equal to 9 vs greater than or equal to 10), proteinuria level (less than 2 g per 24 hr vs greater than or equal to 2 g per 24 hr), and race, and then randomly assigned to receive *BENLYSTA* 1 mg/kg, *BENLYSTA* 10 mg/kg, or placebo in addition to standard of care. The patients were administered study medication intravenously over a 1-hour period on Days 0, 14, 28, and then every 28 days for 48 or 72 weeks.

The primary efficacy endpoint was a composite endpoint (SLE Responder Index) that defined response as meeting each of the following criteria at Week 52 compared with baseline:

- greater than or equal to 4-point reduction in the SELENA-SLEDAI score, and
- no new British Isles Lupus Assessment Group (BILAG) A organ domain score or two new BILAG B organ domain scores, and
- no worsening (less than 0.30 point increase) in Physician's Global Assessment score (PGA).

The SLE Responder Index uses the SELENA-SLEDAI score as an objective measure of reduction in global disease activity; the BILAG index to ensure no significant worsening in any specific organ system; and the PGA to ensure that improvements in disease activity are not achieved at the expense of the patient's overall condition.

BENLYSTA produced significant improvements in the SLE Responder Index as well as in the individual component SELENA-SLEDAI score in both studies, see Table 1.

Table 1: Response Rate at Week 52

Response	Study 1		Study 2		Studies 1 and 2 Pooled	
	Placebo (n=275)	<i>BENLYSTA</i> 10mg/kg (n=273)	Placebo (n=287)	<i>BENLYSTA</i> 10mg/kg (n=290)	Placebo (n=562)	<i>BENLYSTA</i> 10mg/kg (n=563)
SLE Responder Index	33.8%	43.2% (P=0.021)	43.6%	57.6% (P=0.0006)	38.8%	50.6% (P<0.0001)
Components of SLE Responder Index						
Percent of patients with reduction in SELENA-SLEDAI ≥ 4	35.6%	46.9% (P=0.006)	46.0%	58.3% (P= 0.0024)	40.9%	52.8% (P<0.0001)
Percent of patients with no worsening by BILAG index	65.1%	69.2% (P=0.32)	73.2%	81.4% (P=0.018)	69.2%	75.5% (P=0.019)

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Percent of patients with no worsening by PGA	62.9%	69.2% (P=0.13)	69.3%	79.7% (P=0.0048)	66.2%	74.6% (P=0.0017)
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In a pooled analysis of the two studies, the percentage of patients receiving greater than 7.5 mg/ day prednisone (or equivalent) at baseline whose average corticosteroid dose was reduced by at least 25% from baseline to a dose equivalent to prednisone less than or equal to 7.5 mg/day during Weeks 40 through 52, was 17.9% in the group receiving belimumab and 12.3% in the group receiving placebo (P=0.0451).

Flares in SLE were defined by the Modified SELENA-SLEDAI SLE Flare Index where the modification excludes severe flares that are triggered only by an increase of SELENA-SLEDAI score to greater than 12. The median time to the first flare was delayed in the pooled group receiving belimumab compared to the group receiving placebo (hazard ratio=0.84, P=0.012). The risk of severe flares was also reduced by 36% over the 52 weeks of observation in the group receiving belimumab compared to the group receiving placebo (hazard ratio=0.64, P=0.0011).

Univariate and multivariate analysis of the primary endpoint demonstrated that the greatest benefit was observed in patients with higher disease activity at baseline including patients with SELENA-SLEDAI scores greater than or equal to 10, or patients requiring steroids to control their disease, or patients with low complement levels.

Post-hoc analysis identified a high responding subgroup as those patients with low complement and positive anti-dsDNA at baseline, see Table 2 for results of this example of a higher disease activity group. Of these patients, 64.5% had SELENA-SLEDAI scores greater than or equal to 10 at baseline.

Table 2: Patients with low complement and positive anti-dsDNA at baseline

Subgroup	Anti-dsDNA positive AND low complement	
BLISS-76 and BLISS-52 pooled data	<i>Placebo</i> (n=287)	<i>BENLYSTA 10 mg/kg</i> (n=305)
SRI response rate at Week 52 (%)	31.7	51.5 (P<0.0001)
Observed treatment difference vs placebo (%)		19.8
SRI response rate (excluding complement and anti-dsDNA changes) at Week 52 (%)	28.9	46.2 (P<0.0001)
Observed treatment difference vs placebo (%)		17.3

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Severe flares over 52 weeks:		
Patients experiencing a severe flare (%)	29.6	19.0
Observed treatment difference vs placebo (%)		10.6
Time to severe flare [Hazard ratio (95% CI)]		0.61 (0.44, 0.85) (P=0.0038)
Prednisone reduction by $\geq 25\%$ from baseline to ≤ 7.5 mg/day during Weeks 40 through 52* (%)	(n=173) 12.1	(n=195) 18.5 (P=0.0964)
Observed treatment difference vs placebo (%)		6.3
FACIT-fatigue score improvement from baseline at Week-52 (mean):	1.99	4.21 (P=0.0048)
Observed treatment difference vs placebo (mean difference)		2.21
BLISS-76 Study only	Placebo (n=131)	<i>BENLYSTA</i> 10 mg/kg (n=134)
SRI response rate at Week-76 (%):		
Observed treatment difference vs placebo (%)	27.5	39.6 (P=0.0160) 12.1

* Among patients with baseline prednisone dose greater than 7.5 mg/day

Lupus Nephritis

The efficacy and safety of *BENLYSTA* 10 mg/kg administered intravenously over a 1-hour period on Days 0, 14, 28, and then every 28 days, were evaluated in a 104-week randomised (1:1), double-blind, placebo-controlled, Phase III study (BEL114054) in 448 patients with active lupus nephritis. The patients had a clinical diagnosis of SLE according to ACR classification criteria, biopsy proven lupus nephritis Class III, IV, and/or V and had active renal disease at screening requiring standard therapy. Standard therapy included corticosteroids, 0 to 3 intravenous administrations of methylprednisolone (500 to 1000 mg per administration), followed by oral prednisone 0.5 to 1 mg/kg/day with a total daily dose ≤ 60 mg/day and tapered to ≤ 10 mg/day by Week 24, with:

- mycophenolate mofetil 1 to 3 g/day orally or mycophenolate sodium 720 to 2160 mg/day orally for induction and maintenance, or
- cyclophosphamide 500 mg intravenously every 2 weeks for 6 infusions for induction followed by azathioprine orally at a target dose of 2 mg/kg/day for maintenance.

This study was conducted in Asia, North America, South America, and Europe. Patient median age was 31 years (range: 18 to 77 years); the majority (88%) were female.

The primary efficacy endpoint was Primary Efficacy Renal Response (PERR) at Week 104 defined as a response at Week 100 confirmed by a repeat measurement at Week 104 of the following parameters: urinary protein:creatinine ratio (uPCR) ≤ 700 mg/g (79.5 mg/mmol) and estimated glomerular filtration rate (eGFR) ≥ 60 mL/min/1.73m² or no decrease in eGFR of

>20% from pre-flare value.

The major secondary endpoints included:

- Complete Renal Response (CRR) defined as a response at Week 100 confirmed by a repeat measurement at Week 104 of the following parameters: uPCR <500 mg/g (56.8 mg/mmol) and eGFR $\geq$ 90 mL/min/1.73m² or no decrease in eGFR of >10% from pre-flare value.
- PERR at Week 52.
- Time to renal-related event or death (renal-related event defined as first event of end-stage renal disease, doubling of serum creatinine, renal worsening [defined as increased proteinuria, and/or impaired renal function], or receipt of renal disease-related prohibited therapy).

For PERR and CRR endpoints, steroid treatment had to be reduced to $\leq$ 10 mg/day from Week 24 to be considered a responder. For these endpoints, patients who discontinued treatment early, received prohibited medication, or withdrew from the study early were considered non-responders.

The proportion of patients achieving PERR at Week 104 was significantly higher in patients receiving BENLYSTA compared with placebo. The major secondary endpoints also showed significant improvement with BENLYSTA compared with placebo (Table 3).

Table 3: Efficacy results in adult patients with lupus nephritis

Efficacy Endpoint	Placebo n = 223	BENLYSTA 10 mg/kg n = 223	Observed difference vs. placebo	Odds/Hazard ratio vs. placebo (95% CI)	P-value
PERR at Week 104¹ Responders	32.3%	43.0%	10.8%	OR 1.55 (1.04, 2.32)	0.0311
Components of PERR					
Urine protein:creatinine ratio $\leq$ 700 mg/g (79.5 mg/mmol)	33.6%	44.4%	10.8%	OR 1.54 (1.04, 2.29)	0.0320
eGFR $\geq$ 60 mL/min/1.73m ² or no decrease in eGFR from pre-flare value of >20%	50.2%	57.4%	7.2%	OR 1.32 (0.90, 1.94)	0.1599
Not treatment failure ³	74.4%	83.0%	8.5%	OR 1.65 (1.03, 2.63)	0.0364
CRR at Week 104¹ Responders	19.7%	30.0%	10.3%	OR 1.74 (1.11, 2.74)	0.0167
Components of CRR					
Urine protein:creatinine ratio $\leq$ 700 mg/g (56.8 mg/mmol)	28.7%	39.5%	10.8%	OR 1.58 (1.05, 2.38)	0.0268
eGFR $\geq$ 60	39.9%	46.6%	6.7%	OR 1.33	0.1539

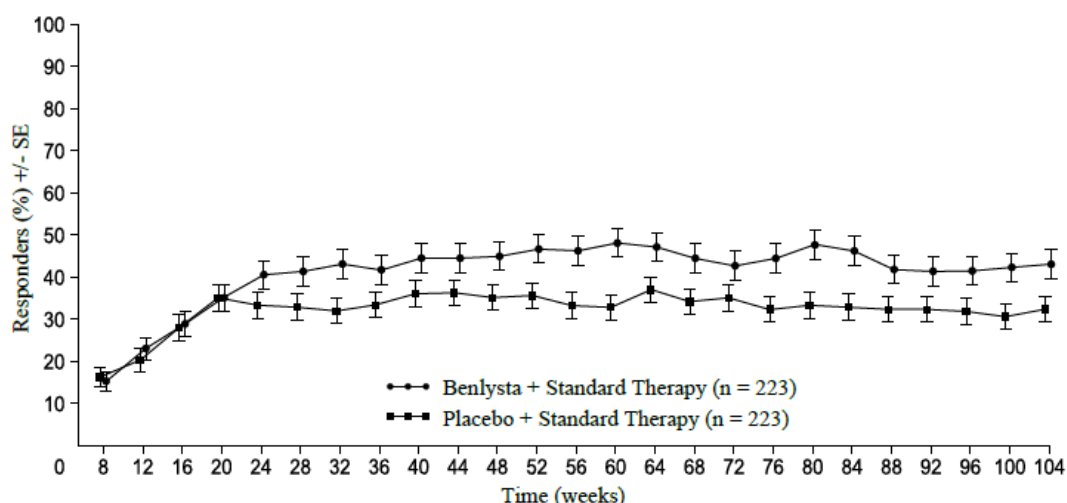
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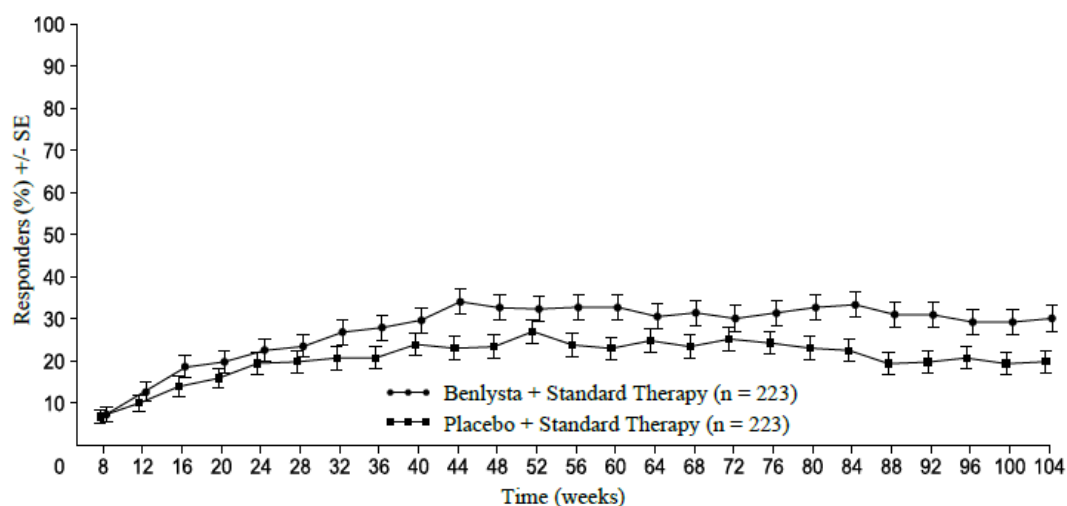
mL/min/1.73m ² or no decrease in eGFR from pre-flare value of >10%				(0.90, 1.96)	
Not treatment failure ³	74.4%	83.0%	8.5%	OR 1.65 (1.03, 2.63)	0.0364
PERR at Week 52¹ Responders	35.4%	46.6%	11.2%	OR 1.59 (1.06, 2.38)	0.0245
Time to Renal- Related Event or Death¹ Percentage of patients with event ²	28.3%	15.7%	-		
Time to event [Hazard ratio (95% CI)]			-	HR 0.51 (0.34, 0.77)	0.0014
¹ PERR at Week 104 was the primary efficacy analysis; CRR at Week 104, PERR at Week 52 and time to renal-related event or death were included in the pre-specified testing hierarchy. ² When excluding deaths from the analysis (1 for BENLYSTA; 2 for placebo), the percentage of patients with a renal-related event was 15.2% for BENLYSTA compared with 27.4% for placebo (HR = 0.51; 95% CI: 0.34, 0.78). ³ Treatment failure: Patients who took protocol-prohibited medication.					

A numerically greater percentage of patients receiving BENLYSTA achieved PERR beginning at Week 24 compared with placebo, and this treatment difference was maintained through to Week 104. Beginning at Week 12, a numerically greater percentage of patients receiving BENLYSTA achieved CRR compared with placebo and the numerical difference was maintained through to Week 104 (Figure 1).

Figure 1: Response Rates in Adults with Lupus Nephritis by Visit

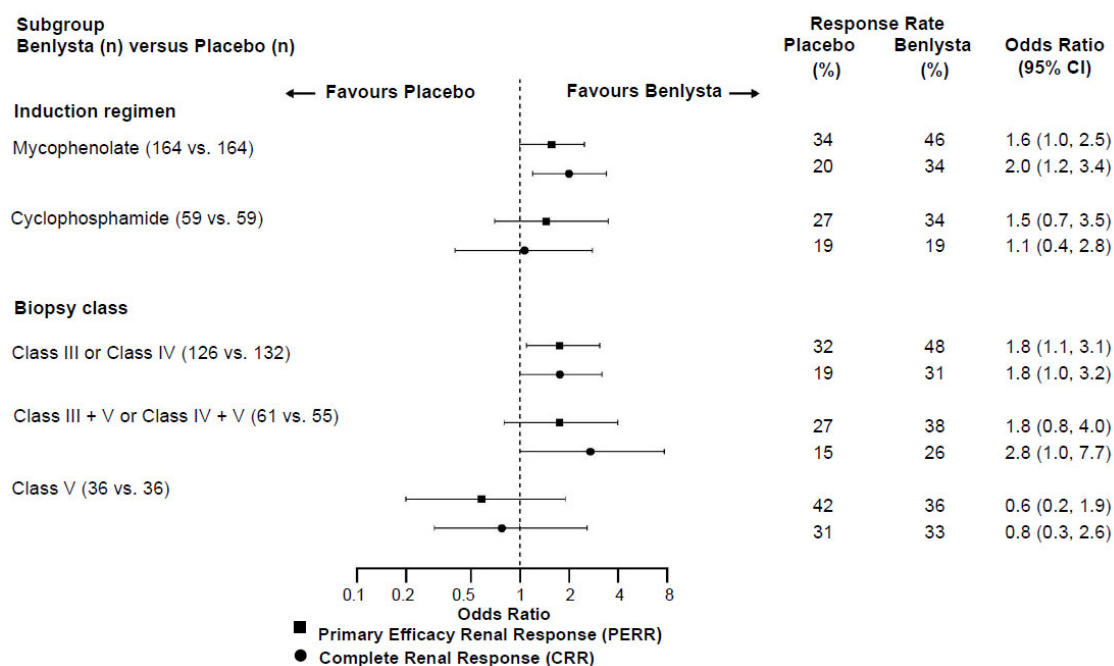
Primary Efficacy Renal Response (PERR)



Complete Renal Response (CRR)

In descriptive subgroup analyses, key efficacy endpoints (PERR and CRR) were examined by induction regimen (mycophenolate or cyclophosphamide) and biopsy class (Class III or IV, Class III + V or Class IV + V, or Class V) (Figure 2).

Figure 2: Odds Ratio of PERR and CRR at Week 104 across Subgroups



Black patients

BENLYSTA was administered intravenously to black patients in a randomised (2:1), double blind, placebo-controlled 52-week Phase III/IV study (BEL115471). Efficacy was evaluated in 448 patients. The study design was the same as the pivotal studies summarised above apart from: eligible patients had a SELENA-SLEDAI score ≥ 8 and the primary endpoint was the SRI response at Week 52 with modified SLEDAI-2K scoring for proteinuria (SRI-S2K). The study was conducted in North America, South America, Europe and Africa. Patient median age was

38 years (range: 18 to 71 years), and the majority of patients (97%) were female.

The proportion of black patients achieving an SRI-S2K response was higher in patients receiving *BENLYSTA* but the difference was not statistically significant compared with placebo. The trends in comparisons between the treatment groups for the rates of response for the individual components of the endpoint were generally consistent with that of the SRI-S2K (Table 4).

Table 4: Response rate in black patients at Week 52

Response	Placebo (n=149)	<i>BENLYSTA</i> 10mg/kg (n=298)
SLE Responder Index (SRI-S2K)	41.6%	48.7%
Odds ratio (95% CI) vs placebo		1.40 (0.93, 2.11) (P=0.1068)
Components of SLE Responder Index (SRI-S2K)		
Percent of patients with reduction in SELENA-SLEDAI-S2K ≥ 4	42.3%	50.0%
Odds ratio (95% CI) vs placebo		1.46 (0.97, 2.20)
Percent of patients with no worsening by BILAG index	62.4%	67.8%
Odds ratio (95% CI) vs placebo		1.24 (0.81, 1.88)
Percent of patients with no worsening by PGA	64.4%	69.5%
Odds ratio (95% CI) vs placebo		1.26 (0.82, 1.93)

¹ Analyses excluded any subject missing a baseline assessment for any of the components (1 for belimumab).

The safety profile of *BENLYSTA* in black patients (n=331) was consistent with the known safety profile of *BENLYSTA* in the overall population.

Consistent with results from other studies, in patients with high disease activity (low complement and positive anti-dsDNA at baseline, n=141) the SRI-S2K response was 45.1% for *BENLYSTA* 10 mg/kg compared with 24.0% for placebo (odds ratio 3.00; 95% CI: 1.35, 6.68). These results suggest a greater relative response to *BENLYSTA* compared to placebo in black patients with high disease activity.

Other special patient groups

There were too few males or patients over 65 years of age enrolled in the controlled clinical trials to draw meaningful conclusions about the effects of gender, age, or race on clinical outcomes.

Non-Clinical Information

Non-clinical data revealed no special hazard for humans based on studies of repeated dose toxicity and toxicity to reproduction.

Intravenous and subcutaneous administration to monkeys resulted in the expected reduction in number of peripheral and lymphoid tissue B-cell counts with no associated toxicological

findings.

Reproductive studies have been performed in pregnant cynomolgus monkeys receiving belimumab 150 mg/kg by intravenous infusion (approximately nine times the anticipated maximum human clinical exposure) every two weeks for up to 21 weeks, and belimumab treatment was not associated with direct or indirect harmful effects with respect to maternal toxicity, developmental toxicity, or teratogenicity. Treatment-related findings were limited to the expected reversible reduction of B cells in both dams and infants and reversible reduction of IgM in infant monkeys. B cell numbers recovered after the cessation of belimumab treatment by about one year post-partum in adult monkeys and by three months of life in infant monkeys; IgM levels in infants exposed to belimumab in utero recovered by six months of age.

As belimumab is a monoclonal antibody, no genotoxicity studies have been conducted. No carcinogenicity or fertility studies (male or female) have been performed.

PHARMACEUTICAL INFORMATION

List of Excipients

Citric acid monohydrate
Sodium citrate dihydrate
Sucrose
Polysorbate 80

Shelf Life

The expiry date is indicated on the packaging.

Storage

The storage conditions are detailed on the packaging.

Unopened vials

Store at between 2°C and 8°C. Do not freeze.

Protect from light. Store in the original carton until use.

Reconstituted solution for intravenous infusion

After reconstitution with Water for Injections, and dilution in 0.9% sodium chloride (normal saline), 0.45% sodium chloride (half normal saline), or Lactated Ringer's solution, the product is stable for up to 8 hours at 2°C to 8°C or at room temperature. Protect from direct sunlight.

Nature and Contents of Container

5 mL Type 1 glass vial sealed with a siliconised rubber stopper and a flip-off seal containing 120 mg *BENLYSTA* as a lyophilised powder.

The drug is supplied in a single use vial without a preservative.

Incompatibilities

BENLYSTA is not compatible with 5% dextrose.

BENLYSTA must be prepared and administered only as directed, see *Use and Handling*.

Use and Handling

Reconstitution and dilution

BENLYSTA does not contain a preservative; therefore reconstitution and dilution must be carried out under aseptic conditions.

Allow 10 to 15 minutes for the vial to warm to room temperature.

It is recommended that a 21-25 gauge needle be used when piercing the vial stopper for reconstitution and dilution.

The 120 mg single-use vial of *BENLYSTA* should be reconstituted with 1.5 mL of sterile Water for Injections to yield a final concentration of 80 mg/mL belimumab. The 400 mg single-use vial of *BENLYSTA* should be reconstituted with 4.8 mL of sterile Water for Injections to yield a final concentration of 80 mg/mL belimumab.

The stream of sterile water should be directed toward the side of the vial to minimise foaming. Gently swirl the vial for 60 seconds. Allow the vial to sit at room temperature during reconstitution, gently swirling the vial for 60 seconds every five minutes until the powder is dissolved. Do not shake.

Reconstitution is typically complete within 10 to 15 minutes after the sterile water has been added, but it may take up to 30 minutes. Protect the reconstituted solution from direct sunlight.

If a mechanical reconstitution device is used to reconstitute *BENLYSTA* it should not exceed 500 rpm and the vial should be swirled for no longer than 30 minutes.

Once reconstitution is complete, the solution should be opalescent and colourless to pale yellow, and without particles. Small air bubbles, however, are expected and acceptable.

The reconstituted product is diluted to 250 mL with 0.9% sodium chloride (normal saline), 0.45% sodium chloride (half normal saline) or Lactated Ringer's solution for intravenous infusion. For patients whose body weight is less than or equal to 40 kg, infusion bags with 100 mL normal saline, half normal saline or Lactated Ringer's solution may be considered providing that the resulting belimumab concentration in the infusion bag does not exceed 4 mg/mL.

5% Dextrose intravenous solutions are incompatible with *BENLYSTA* and should not be used.

From a 250 mL (or 100 mL) infusion bag or bottle of normal saline, half normal saline, or Lactated Ringer's solution, withdraw and discard a volume equal to the volume of the reconstituted *BENLYSTA* solution required for the patient's dose. Then add the required volume of the reconstituted *BENLYSTA* solution into the infusion bag or bottle. Gently invert the bag or bottle to mix the solution. Any unused solution in the vials must be discarded.

Inspect the *BENLYSTA* solution visually for particulate matter and discoloration prior to administration. Discard the solution if any particulate matter or discoloration is observed.

The reconstituted solution, if not used immediately, should be protected from direct sunlight and stored refrigerated at 2°C to 8°C. Solutions diluted in normal saline, half normal saline, or Lactated Ringer's solution may be stored at 2°C to 8°C or room temperature.

The total time from reconstitution of *BENLYSTA* to completion of infusion should not exceed

eight hours.

Administration

BENLYSTA should be infused over a 1-hour period.

BENLYSTA should not be infused concomitantly in the same intravenous line with other agents. No physical or biochemical compatibility studies have been conducted to evaluate the co-administration of *BENLYSTA* with other agents.

No incompatibilities between *BENLYSTA* and polyvinylchloride or polyolefin bags have been observed.

Not all presentations are available in every country.

Manufacturer

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Version number: 08

Reference: GDS23/IPI20 & EU SmPC

Date of local revision: 11 September 2025